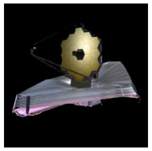

9111 - An Infrared View of the Type Iax Supernova Remnant Pa 30

Cycle: 4, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Ilaria Caiazzo (PI) (ESA Member)	**Institute of Science and Technology Austria**
Dr. Tim Cunningham (CoI) (CoPI)	Harvard University
Philippe Z. Yao (CoI)	Princeton University
Mr. Nikolaus Zen Prusinski (CoI)	California Institute of Technology
Prof. Jim Fuller (CoI)	California Institute of Technology
Prof. Shrinivas R. Kulkarni (CoI)	California Institute of Technology
Dr. Scott J. Kenyon (CoI)	Smithsonian Institution Astrophysical Observatory
Dr. John Charles Raymond (CoI)	Smithsonian Institution Astrophysical Observatory
Prof. Abigail Polin (CoI)	Purdue University
Prof. Eliot Quataert (CoI)	Princeton University
Prof. Paul Duffell (CoI)	Purdue University
Prof. David Charbonneau (CoI) (US Admin CoI)	Harvard University

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Science				
	1	Science 1	MIRI Medium Resolution Spectroscopy	(1) Pa30-1
	2	Science 2	MIRI Medium Resolution Spectroscopy	(2) Pa30-2
	3	Science 3	MIRI Medium Resolution Spectroscopy	(3) Pa30-3
	4	Science 4	MIRI Medium Resolution Spectroscopy	(4) Pa30-4
	5	Science 5	MIRI Medium Resolution Spectroscopy	(5) Pa30-5
	6	Science 6	MIRI Medium Resolution Spectroscopy	(7) Pa30-6
Background				

Folder	Observation	Label	Observing Template	Science Target
	7	Background 1	MIRI Medium Resolution Spectroscopy	(6) Pa30-bkg
	8	Background 2	MIRI Medium Resolution Spectroscopy	(6) Pa30-bkg
	9	Background 3	MIRI Medium Resolution Spectroscopy	(6) Pa30-bkg
	10	Background 4	MIRI Medium Resolution Spectroscopy	(6) Pa30-bkg
	11	Background 5	MIRI Medium Resolution Spectroscopy	(6) Pa30-bkg
	12	Background 6	MIRI Medium Resolution Spectroscopy	(6) Pa30-bkg

ABSTRACT

Pa 30 is an extraordinary and unique supernova remnant. A surviving hot star is at its center and shows strong outflows with extreme velocities in excess of 15,000 km/s. Optical spectra of the central star and the nebula lack any indication of hydrogen or helium, while an X-ray spectrum reveals carbon-burning ashes. But the most striking feature is the morphology of the nebula, unlike any previously seen in a supernova remnant: the optical emission is structured in almost radial filaments distributed in a spherically symmetric fashion around the central star. Because of the surviving stellar remnant and the lack of hydrogen and helium, it has been suggested that Pa 30 is the product of a thermonuclear explosion in a near- or super-Chandrasekhar white dwarf that failed to explode the entire star and created a sub-luminous transient. These failed explosions have been observed in other galaxies and are considered a sub-class of type Ia supernovae called type Iax. The discovery of Pa 30 gives us the unique opportunity to study the physics and evolution of the remnant of such an event in our own Galaxy. We here propose to observe 6 sections of the nebula with MIRI/MRS, and the entire nebula with MIRI imaging to 1) reveal if the IR emission is diffuse or filamentary in nature, 2) constrain the contribution of thermal dust and line-emission in the IR, and 3) characterize the physical properties (composition, density, temperature) of the nebula by comparing lines in the IR and in the optical. With its unprecedented angular resolution and sensitivity in the mid-IR, JWST is uniquely suited for this study, which will be key to understand the origin and evolution of this unique SNR.

OBSERVING DESCRIPTION

We propose to observe with MIRI/IFU 6 bright filaments in the nebula, 4 of which overlap with existing optical IFU observations of bright filaments, to obtain medium resolution spectra and detect line emission across the nebula. Taking advantage of the simultaneous MIRI imaging, we also propose to obtain an image of the entire nebula in three filters: F770W, F1280W and F2550W.

For each science pointing, we request 37 minutes of exposure time in each of the three MIRI/MRS wavelength ranges and each of the three imaging filters. We request specific PA ranges so that we can obtain images of the entire nebula in the three filters. Additionally, since the nebula extends throughout the entire field of view of our observations, we also request an off-side background pointing for each science pointing, imposing the "non interruptible" requirement . This sums up to a total exposure time of 13.6 hours (26.7 with overheads). We use the optimized 4-point dithering pattern for extended sources for our science observations, and the 2-point pattern optimized for background for our background pointings. We do not request target acquisition because the 0.3" pointing accuracy is enough for our purpose.

From the images, we will be able to determine the fraction of diffuse IR emission, constrain the contribution and temperature of the dust bump and its variation across the nebula, especially its spatial correlation with the filaments. Also, we will locate with much higher angular accuracy the edge between the inner bright ring and the outer IR halo. From the IFU spectra, we expect to detect several emission lines, which, when combined with the optical IFU data, will constrain the density, ionization state and composition of the filaments, as well as the velocity of the ejecta. The FOV of the IFU is large enough for us to also analyze the spatial distribution and intensity of line emission in and outside the filaments. We will therefore obtain a much stronger constraint on the density and filling factor of the filaments than from optical spectroscopy alone, which will allow us to calculate the total ejecta mass. By targeting filaments in different regions of the infrared nebula, the combination of the IFU with the imaging will allow us to constrain the contribution of line vs dust emission, and therefore constrain the dust-to-gas ratio across the nebula.

Proposal 9111 - Targets - An Infrared View of the Type Iax Supernova Remnant Pa 30

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Pa30-1	RA: 00 53 11.0580 (13.2960750d) Dec: +67 29 49.43 (67.49706d) Equinox: J2000		
Comments: Category=Star Description=[Supernovae]				
(2)	Pa30-2	RA: 00 53 25.0000 (13.3541667d) Dec: +67 30 7.58 (67.50211d) Equinox: J2000		
Comments: Category=Star Description=[Supernovae]				
(3)	Pa30-3	RA: 00 53 20.2600 (13.3344167d) Dec: +67 29 52.54 (67.49793d) Equinox: J2000		
Comments: Category=Star Description=[Supernovae]				
(4)	Pa30-4	RA: 00 53 30.3457 (13.3764404d) Dec: +67 29 51.30 (67.49758d) Equinox: J2000		
Comments: Category=Star Description=[Supernovae]				
(5)	Pa30-5	RA: 00 53 13.7376 (13.3072400d) Dec: +67 29 59.28 (67.49980d) Equinox: J2000		
Comments: Category=Star Description=[Supernovae]				
(6)	Pa30-bkg	RA: 00 51 54.3695 (12.9765396d) Dec: +67 30 1.58 (67.50044d) Equinox: J2000		
Comments: Category=Calibration Description=[Telescope/sky background]				
(7)	Pa30-6	RA: 00 53 21.3739 (13.3390579d) Dec: +67 30 47.02 (67.51306d) Equinox: J2000		
Comments: Category=Star Description=[Supernovae]				

Observation

Proposal 9111, Observation 1: Science 1

Thu Aug 28 17:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Background 1 (Obs 7)]

Diagnostics

(Science 1 (Obs 1)) Warning (Form): Imager Filter overlap.

(Science 1 (Obs 1)) Warning (Form): Imager Filter overlap.

(Visit 1:1) Warning (Form): Data Excess over lower threshold

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	Pa30-1	RA: 00 53 11.0580 (13.2960750d) Dec: +67 29 49.43 (67.49706d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	100	2	1	Dither 1	4	8	2231.132	221056.17
1	SHORT(A)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
1	SHORT(A)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	
2		IMAGER	F1280W	FASTR1	100	2	1	Dither 1	4	8	2231.132	221056.20
2	MEDIUM(B)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
2	MEDIUM(B)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3		IMAGER	F2550W	FASTR1	15	12	1	Dither 1	4	48	2120.131	221056.24
3	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	

Special Requirements	
	Aperture PA Range 165 to 175 Degrees (V3 165.0 to 175.0) Group Observations 1, 7, Non-interruptible

Observation

Proposal 9111, Observation 2: Science 2

Thu Aug 28 17:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Background 2 (Obs 8)]

Diagnostics

(Science 2 (Obs 2)) Warning (Form): Imager Filter overlap.

(Science 2 (Obs 2)) Warning (Form): Imager Filter overlap.

(Visit 2:1) Warning (Form): Data Excess over lower threshold

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	Pa30-2	RA: 00 53 25.0000 (13.3541667d) Dec: +67 30 7.58 (67.50211d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	100	2	1	Dither 1	4	8	2231.132	221056.17
1	SHORT(A)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
1	SHORT(A)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	
2		IMAGER	F1280W	FASTR1	100	2	1	Dither 1	4	8	2231.132	221056.20
2	MEDIUM(B)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
2	MEDIUM(B)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3		IMAGER	F2550W	FASTR1	15	12	1	Dither 1	4	48	2120.131	221056.24
3	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	

Special Requirements	
	Aperture PA Range 80 to 90 Degrees (V3 80.0 to 90.0) Group Observations 2, 8, Non-interruptible

Observation

Proposal 9111, Observation 3: Science 3

Thu Aug 28 17:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Background 3 (Obs 9)]

Diagnostics

(Science 3 (Obs 3)) Warning (Form): Imager Filter overlap.
(Science 3 (Obs 3)) Warning (Form): Imager Filter overlap.
(Visit 3:1) Warning (Form): Data Excess over lower threshold
(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	Pa30-3	RA: 00 53 20.2600 (13.3344167d) Dec: +67 29 52.54 (67.49793d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	100	2	1	Dither 1	4	8	2231.132	221056.17
1	SHORT(A)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
1	SHORT(A)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	
2		IMAGER	F1280W	FASTR1	100	2	1	Dither 1	4	8	2231.132	221056.20
2	MEDIUM(B)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
2	MEDIUM(B)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3		IMAGER	F2550W	FASTR1	15	12	1	Dither 1	4	48	2120.131	221056.24
3	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	

Special Requirements	
	Aperture PA Range 165 to 175 Degrees (V3 165.0 to 175.0) Group Observations 3, 9, Non-interruptible

Observation

Proposal 9111, Observation 4: Science 4

Thu Aug 28 17:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Background 4 (Obs 10)]

Diagnostics

(Science 4 (Obs 4)) Warning (Form): Imager Filter overlap.

(Science 4 (Obs 4)) Warning (Form): Imager Filter overlap.

(Visit 4:1) Warning (Form): Data Excess over lower threshold

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	Pa30-4	RA: 00 53 30.3457 (13.3764404d) Dec: +67 29 51.30 (67.49758d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	100	2	1	Dither 1	4	8	2231.132	221056.17
1	SHORT(A)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
1	SHORT(A)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	
2		IMAGER	F1280W	FASTR1	100	2	1	Dither 1	4	8	2231.132	221056.20
2	MEDIUM(B)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
2	MEDIUM(B)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3		IMAGER	F2550W	FASTR1	15	12	1	Dither 1	4	48	2120.131	221056.24
3	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	

Special Requirements	
	Aperture PA Range 165 to 175 Degrees (V3 165.0 to 175.0) Group Observations 4, 10, Non-interruptible

Observation

Proposal 9111, Observation 5: Science 5

Thu Aug 28 17:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Background 5 (Obs 11)]

Diagnostics

(Science 5 (Obs 5)) Warning (Form): Imager Filter overlap.

(Science 5 (Obs 5)) Warning (Form): Imager Filter overlap.

(Visit 5:1) Warning (Form): Data Excess over lower threshold

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(5)	Pa30-5	RA: 00 53 13.7376 (13.3072400d) Dec: +67 29 59.28 (67.49980d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	100	2	1	Dither 1	4	8	2231.132	221056.17
1	SHORT(A)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
1	SHORT(A)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	
2		IMAGER	F1280W	FASTR1	100	2	1	Dither 1	4	8	2231.132	221056.20
2	MEDIUM(B)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
2	MEDIUM(B)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3		IMAGER	F2550W	FASTR1	15	12	1	Dither 1	4	48	2120.131	221056.24
3	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	

Special Requirements	
	Aperture PA Range 80 to 90 Degrees (V3 80.0 to 90.0) Group Observations 5, 11, Non-interruptible

Observation

Proposal 9111, Observation 6: Science 6

Thu Aug 28 17:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[Background 6 (Obs 12)]

Diagnostics

(Science 6 (Obs 6)) Warning (Form): Imager Filter overlap.

(Science 6 (Obs 6)) Warning (Form): Imager Filter overlap.

(Visit 6:1) Warning (Form): Data Excess over lower threshold

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(7)	Pa30-6	RA: 00 53 21.3739 (13.3390579d) Dec: +67 30 47.02 (67.51306d) Equinox: J2000		

Comments:
Category=Star
Description=[Supernovae]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	100	2	1	Dither 1	4	8	2231.132	221056.17
1	SHORT(A)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
1	SHORT(A)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	
2		IMAGER	F1280W	FASTR1	100	2	1	Dither 1	4	8	2231.132	221056.20
2	MEDIUM(B)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
2	MEDIUM(B)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3		IMAGER	F2550W	FASTR1	15	12	1	Dither 1	4	48	2120.131	221056.24
3	LONG(C)	MRSLONG		FASTR1	100	2	1	Dither 1	4	8	2231.132	
3	LONG(C)	MRSSHORT		FASTR1	100	2	1	Dither 1	4	8	2231.132	

Special Requirements	
	Aperture PA Range 10 to 40 Degrees (V3 10.0 to 40.0) Group Observations 6, 12, Non-interruptible

Observation

Proposal 9111, Observation 7: Background 1

Thu Aug 28 17:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Science 1 (Obs 1)]

Diagnostics

(Background 1 (Obs 7)) Warning (Form): Imager Filter overlap.

(Background 1 (Obs 7)) Warning (Form): Imager Filter overlap.

(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	Pa30-bkg	RA: 00 51 54.3695 (12.9765396d) Dec: +67 30 1.58 (67.50044d) Equinox: J2000		

Comments:
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	2-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	100	1	1	Dither 1	2	2	555.008	221056.29
1	SHORT(A)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
1	SHORT(A)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	
2		IMAGER	F1280W	FASTR1	100	1	1	Dither 1	2	2	555.008	
2	MEDIUM(B)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
2	MEDIUM(B)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	
3		IMAGER	F2550W	FASTR1	15	6	1	Dither 1	2	12	527.258	
3	LONG(C)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
3	LONG(C)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	

Special Requirements	Group Observations 1, 7, Non-interruptible

Observation

Proposal 9111, Observation 8: Background 2

Thu Aug 28 17:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Science 2 (Obs 2)]

Diagnostics

(Background 2 (Obs 8)) Warning (Form): Imager Filter overlap.

(Background 2 (Obs 8)) Warning (Form): Imager Filter overlap.

(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	Pa30-bkg	RA: 00 51 54.3695 (12.9765396d) Dec: +67 30 1.58 (67.50044d) Equinox: J2000		

Comments:
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	2-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	100	1	1	Dither 1	2	2	555.008	221056.29
1	SHORT(A)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
1	SHORT(A)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	
2		IMAGER	F1280W	FASTR1	100	1	1	Dither 1	2	2	555.008	
2	MEDIUM(B)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
2	MEDIUM(B)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	
3		IMAGER	F2550W	FASTR1	15	6	1	Dither 1	2	12	527.258	
3	LONG(C)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
3	LONG(C)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	

Special Requirements	
	Group Observations 2, 8, Non-interruptible

Observation

Proposal 9111, Observation 9: Background 3

Thu Aug 28 17:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Science 3 (Obs 3)]

Diagnostics

(Background 3 (Obs 9)) Warning (Form): Imager Filter overlap.

(Background 3 (Obs 9)) Warning (Form): Imager Filter overlap.

(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	Pa30-bkg	RA: 00 51 54.3695 (12.9765396d) Dec: +67 30 1.58 (67.50044d) Equinox: J2000		

Comments:
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	2-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	100	1	1	Dither 1	2	2	555.008	221056.29
1	SHORT(A)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
1	SHORT(A)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	
2		IMAGER	F1280W	FASTR1	100	1	1	Dither 1	2	2	555.008	
2	MEDIUM(B)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
2	MEDIUM(B)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	
3		IMAGER	F2550W	FASTR1	15	6	1	Dither 1	2	12	527.258	
3	LONG(C)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
3	LONG(C)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	

Special Requirements	
	Group Observations 3, 9, Non-interruptible

Observation

Proposal 9111, Observation 10: Background 4

Thu Aug 28 17:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Science 4 (Obs 4)]

Diagnostics

(Background 4 (Obs 10)) Warning (Form): Imager Filter overlap.

(Background 4 (Obs 10)) Warning (Form): Imager Filter overlap.

(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	Pa30-bkg	RA: 00 51 54.3695 (12.9765396d) Dec: +67 30 1.58 (67.50044d) Equinox: J2000		

Comments:
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	2-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	100	1	1	Dither 1	2	2	555.008	221056.29
1	SHORT(A)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
1	SHORT(A)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	
2		IMAGER	F1280W	FASTR1	100	1	1	Dither 1	2	2	555.008	
2	MEDIUM(B)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
2	MEDIUM(B)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	
3		IMAGER	F2550W	FASTR1	15	6	1	Dither 1	2	12	527.258	
3	LONG(C)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
3	LONG(C)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	

Special Requirements	Group Observations 4, 10, Non-interruptible

Observation

Proposal 9111, Observation 11: Background 5

Thu Aug 28 17:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Science 5 (Obs 5)]

Diagnostics

(Background 5 (Obs 11)) Warning (Form): Imager Filter overlap.

(Background 5 (Obs 11)) Warning (Form): Imager Filter overlap.

(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	Pa30-bkg	RA: 00 51 54.3695 (12.9765396d) Dec: +67 30 1.58 (67.50044d) Equinox: J2000		

Comments:
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	2-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	100	1	1	Dither 1	2	2	555.008	221056.29
1	SHORT(A)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
1	SHORT(A)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	
2		IMAGER	F1280W	FASTR1	100	1	1	Dither 1	2	2	555.008	
2	MEDIUM(B)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
2	MEDIUM(B)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	
3		IMAGER	F2550W	FASTR1	15	6	1	Dither 1	2	12	527.258	
3	LONG(C)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
3	LONG(C)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	

Special Requirements	
	Group Observations 5, 11, Non-interruptible

Observation

Proposal 9111, Observation 12: Background 6

Thu Aug 28 17:00:09 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [Science 6 (Obs 6)]

Diagnostics

(Background 6 (Obs 12)) Warning (Form): Imager Filter overlap.

(Background 6 (Obs 12)) Warning (Form): Imager Filter overlap.

(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(6)	Pa30-bkg	RA: 00 51 54.3695 (12.9765396d) Dec: +67 30 1.58 (67.50044d) Equinox: J2000		

Comments:
Category=Calibration
Description=[Telescope/sky background]

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	2-Point	BACKGROUND	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1		IMAGER	F770W	FASTR1	100	1	1	Dither 1	2	2	555.008	221056.29
1	SHORT(A)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
1	SHORT(A)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	
2		IMAGER	F1280W	FASTR1	100	1	1	Dither 1	2	2	555.008	
2	MEDIUM(B)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
2	MEDIUM(B)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	
3		IMAGER	F2550W	FASTR1	15	6	1	Dither 1	2	12	527.258	
3	LONG(C)	MRSLONG		FASTR1	100	1	1	Dither 1	2	2	555.008	
3	LONG(C)	MRSSHORT		FASTR1	100	1	1	Dither 1	2	2	555.008	

Special Requirements	Group Observations 6, 12, Non-interruptible